

MODEL SELECTION, PREDICTION AND ESTIMATION FOR MULTIVARIATE NORMAL POPULATIONS

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SUMMARY

The prediction of future observations from multivariate normal populations have been considered using Bayesian techniques and predictive sample reuse methods. Selecting the model that yields the best prediction was considered and simulations comparing different predictors and estimators were made.

1. Introduction

Let $X^{(N)} = \{X_j^{(i)}; j = 1, \dots, N_i; i = 1, \dots, q\}$ denote a random sample of

$$N = N_1 + \dots + N_q$$

observations from q k -dimensional multivariate normal populations with means $m^{(i)}$, $i = 1, \dots, q$ and precision matrix r . Assuming an improper prior joint density on the parameters $M^{(i)}$, $i = 1, \dots, q$ and R , i.e. $\xi(m^{(i)}, r) = |r|^{-\frac{1}{2}(k+1)}$, (see De Groot (1970), Geisser (1964), (1965)) it can be shown that the predictive density function of a future observation,

$$Z = \left(\sqrt{\frac{N_1}{N_1 + 1}} X_{N_1 + 1}^{(1)}, \dots, \sqrt{\frac{N_q}{N_q + 1}} X_{N_q + 1}^{(q)} \right), \text{ where } X_{N_i + 1}^{(i)}$$

denotes the new observation from the i -th population, is given by

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$$(1.1) \quad P_{\tilde{Z}}(Z|x^{(N)}) = c |I_q + (Z-W)' S_N^{-1} (Z-W)|^{-\frac{1}{2}(N+q-1)}, \quad -\infty < Z < \infty, \\ q \leq k,$$

where

$$W = \left(\sqrt{\frac{N_1}{N_1+1}} \bar{x}^{(1)}, \dots, \sqrt{\frac{N_q}{N_q+1}} \bar{x}^{(q)} \right),$$

$$\bar{x}^{(i)} = \frac{1}{N_i} \sum_{j=1}^{N_i} x_j^{(i)},$$

$$S_N = \sum_{i=1}^q \sum_{j=1}^{N_i} (x_j^{(i)} - \bar{x}^{(i)}) (x_j^{(i)} - \bar{x}^{(i)})'$$

and the normalizing constant c is

$$c = \frac{\Gamma_q(\frac{1}{2}(N+q-1))}{\Gamma_q(\frac{1}{2}(N-k+q-1))} \pi^{-\frac{1}{2}kq} |S_N|^{-\frac{1}{2}q}$$

(1.1) is called, according to Geisser (1965), a determinantal distribution with mean W , covariance $S_N/(N+q-k-1)$ and $N+q-k-1$ degrees of freedom. It can be shown that if $Z = (Z_1', Z_2')$, where $Z_1 (k \times q_1)$, $q = q_1 + q_2$, then the marginal distribution of Z_1 will again be a determinantal distribution with mean $W_1 (k \times q_1)$, covariance $S_N/(N+q_1-k-1)$ and $N+q_1-k-1$ degrees of freedom. Using the mean of (1.1) as a predictor of a future observation, it follows that the predictor of X_{N_i+1} is

$$(1.2) \quad E(X_{N_i+1}^{(i)} | x^{(N)}) = \bar{x}_{N_i}^{(i)}, \quad i = 1, \dots, q.$$

One can think of course of many other predictors and this question of prediction and estimation will be considered in more detail in section 3. Before embarking on this however, the problem of deciding how many populations have been sampled from, will first be looked into from the prediction point of view. This will be considered in section 2 under the heading of model selection.

2. Sample reuse criteria for model selection in more than two populations

Geisser (1980 a, b) considered three criteria for selection and allocation in two populations Π_1 and Π_2 , where $\Pi_i \equiv N(m^{(i)}, r^{-1})$, through the method of sample reuse. In this section we shall look into the problem of extending two of these criteria to situa-

tions of more than two populations $\Pi_1, \dots, \Pi_q$.

- (i) The first method which Geisser called the predictive sample reuse (PSR) quasi-Bayes technique approximates the posterior probability assuming the source Π_i of the omitted observation known. It is based on selecting κ such that the likelihood

$$(2.1) \quad L(\omega_\kappa) = \prod_{i=1}^q \prod_{j=1}^{N_i} P_{X_j^{(i)}}(x_j^{(i)} | x^{(N-1)}, \omega_\kappa), \kappa = 1, \dots, m$$

is a maximum. $x^{(N-1)}$ denotes the sample, but omitting the observation $x_j^{(i)}$.

$\omega_\kappa \in \{\omega_1, \dots, \omega_m\}$ is the κ -th model, where for instance ω_1 can be the model specifying

$$\omega_1: m(1) = m(2) = \dots = m(q)$$

among the populations specified in section 1. $P_{X_j^{(i)}}(\cdot)$ is the predictive density of $X_j^{(i)}$ given $x^{(N-1)}$ and ω_κ . From (1.1) it follows that the marginal predictive density function of $X_j^{(i)}$ given $x^{(N-1)}$ is given by:

$$(2.2) \quad P_{X_j^{(i)}}(x_j^{(i)} | x^{(N-1)}, \omega_1) = \frac{\Gamma(\frac{1}{2}(N-1))}{\Gamma(\frac{1}{2}(N-k-1))} \pi^{-\frac{1}{2}k} |S_{N-1}^{[i]}(j)|^{-\frac{1}{2}} \left(\frac{N_i-1}{N_i}\right)^{\frac{1}{2}k}$$

$$\left[1 + \frac{N_i-1}{N_i} (x_j^{(i)} - \bar{x}_j^{[i]})' (S_{N-1}^{[i]}(j))^{-1} (x_j^{(i)} - \bar{x}_j^{[i]}) \right]^{-\frac{1}{2}(N-1)},$$

$$\text{where } S_{N-1}^{[i]}(j) = \sum_{t=1}^q \sum_{\substack{k=1 \\ k \neq j(i)}}^{N_t} (x_k^{(t)} - \bar{x}_j^{[i]})(x_k^{(t)} - \bar{x}_j^{[i]})',$$

$$\bar{x}_j^{[i]} = \frac{1}{N-1} \sum_{t=1}^q \sum_{\substack{k=1 \\ k \neq j(i)}}^{N_t} x_k^{(t)}$$

and $k \neq j(i)$ implies $k \neq j$ if $t=i$.

Hence $X_j^{(i)}$ given $x^{(N-1)}$ and the model ω_1 is distributed as a multivariate t distribution with mean $\bar{x}_j^{[i]}$, covariance $\frac{N_i}{(N_i-1)(N-1-k)} S_{N-1}^{[i]}(j)$ and $N-1-K$ degrees of freedom.

Substituting (2.2) in (2.1) the likelihood can be computed. Similarly $L(\omega_\kappa)$ correspond-

ing to the other models ω_{κ} , $\kappa=2, \dots, m$ can be computed.

Example 2.1

As an example of this procedure a simulation of $N_1 = 10$, $N_2 = 15$ and $N_3 = 20$ observations from $N(m^{(i)}, I_3)$, $i=1, 2, 3$ populations respectively has been done. The values for $m^{(i)}$ considered were

$$m^{(1)} = \begin{bmatrix} 2,70 \\ 4,15 \\ 4,97 \end{bmatrix}, \quad m^{(2)} = \begin{bmatrix} 5,47 \\ 9,57 \\ 19,51 \end{bmatrix}, \quad m^{(3)} = \begin{bmatrix} -1,80 \\ -5,29 \\ -10,17 \end{bmatrix}$$

$$\begin{aligned} \text{Let } \omega_1 : m^{(1)} &= m^{(2)} = m^{(3)} \\ \omega_2 : m^{(1)} &\neq m^{(2)} = m^{(3)} \\ \omega_3 : m^{(1)} &= m^{(3)} \neq m^{(2)} \\ \omega_4 : m^{(1)} &= m^{(2)} \neq m^{(3)} \\ \omega_5 : m^{(1)} &\neq m^{(2)} \neq m^{(3)} \end{aligned}$$

It has been calculated that

$$\begin{aligned} \log L(\omega_1) &= -0,6075(10^4), \log L(\omega_2) = -0,5768(10^4), \\ \log L(\omega_3) &= -0,5273(10^4), \log L(\omega_4) = -0,4961(10^4), \\ \log L(\omega_5) &= -0,3785(10^4). \quad (\text{The calculations were carried out using double precision.}) \end{aligned}$$

Hence we conclude that model ω_5 is appropriate for predicting future observations.

The same example has been considered with $m^{(2)'} = m^{(1)'} = (2,70; 4,15; 4,97)$ and the following log likelihoods were obtained.

$$\begin{aligned} \log L(\omega_1) &= -0,5538(10^4), \log L(\omega_2) = -0,5082(10^4), \\ \log L(\omega_3) &= -0,4904(10^4), \log L(\omega_4) = -0,3534(10^4), \\ \log L(\omega_5) &= -0,3549(10^4). \end{aligned}$$

Hence we conclude that model ω_4 is the appropriate model.

- (ii) The next method, based on the method by Geisser (1980 b) for selection in two populations, treats the omitted observation as if its source is unknown and proceeds to simulate the classification scheme itself by assigning each omitted observation to Π_1 or Π_2 and then determines the number correctly classified for each ω_{κ} . The me-

thod is to calculate

$$(2.3) \quad R_{ij}^t(\omega_\kappa) = \frac{P_{X_j^{(i)}}(x_j^{(i)} | x^{(N-1)}, \Pi_i, \omega_\kappa)}{P_{X_j^{(i)}}(x_j^{(i)} | x^{(N-1)}, \Pi_t, \omega_\kappa)} \quad t \neq i$$

for all $i = 1, \dots, q; j = 1, \dots, N_j; \kappa = 1, \dots, m$.

If $R_{ij}^t(\omega_\kappa) \geq 1$, $x_j^{(i)}$ is assigned to Π_i , otherwise to Π_t . Let $n_i(\omega_\kappa)$ be the number of correct classifications to the i -th population under model ω_κ . Determine κ such that

(2.4) $Q(\omega_\kappa) = n_1(\omega_\kappa) + \dots + n_q(\omega_\kappa)$ is a maximum. It is easy to check that

$$Q(\omega_\kappa) \leq 2(N_1 + N_2 + \dots + N_q).$$

Example 2.2

The first case considered in example (2.1) is used to calculate (2.4). It has been calculated that $Q(\omega_1) = 61$, $Q(\omega_2) = 66$, $Q(\omega_3) = 70$, $Q(\omega_4) = 79$, $Q(\omega_5) = 86$. Hence we conclude that $\kappa=5$, which is the same result as found as in example 2.1 and which gives of course the true ω_5 . The other method suggested by Geisser is not generalized here.

3. Prediction and estimation

We have seen in section 1, equation 1.2 that in the case of improper prior densities on $M^{(i)}$ and R and because of unequal sample sizes, the mean of the predictive density reduces to the sample mean. Different types of priors will however give different predictions.

In this case we are going to assume R fixed, say $R^{-1} = \Sigma$, equal sample sizes $N_i = \bar{N}$ and $M^{(i)} \sim N(\mu, B)$ ($i = 1, \dots, q$).

It is therefore true that

$$X_j^{(i)} | M^{(i)} = m^{(i)} \sim N(m^{(i)}, \Sigma) \quad (i=1, \dots, q)$$

and
$$\bar{X}^{(i)} | M^{(i)} = m^{(i)} \sim N(m^{(i)}, \frac{1}{\bar{N}} \Sigma)$$

where
$$\bar{X}^{(i)} = \frac{1}{\bar{N}} \sum_{j=1}^{\bar{N}} X_j^{(i)}.$$

The posterior distribution of $M^{(i)}$ is given by the conditional distribution

$$M^{(i)} | \bar{X}^{(i)} = \bar{x}^{(i)} \sim N(\mu^{*(i)}, \Sigma^*) \quad (i=1, \dots, q),$$

$$(3.1) \text{ where } \mu^{*(i)} = (I-A) \bar{x}^{(i)} + A \mu,$$

$$\Sigma^* = \frac{1}{\bar{N}} (I-A) \Sigma,$$

$$\text{and } A = \frac{1}{\bar{N}} \Sigma (B + \frac{1}{\bar{N}} \Sigma)^{-1}.$$

It is known for the case of normal populations that the mean $\mu^{*(i)}$ of the posterior distribution is also the mean of the predictive density function (Geisser (1964)). If B , Σ and μ are known then $\mu^{*(i)}$ can be used for the estimation of $m^{(i)}$ as well as for the prediction of a future observation. Usually the parameters are unknown and have to be estimated. As an estimate of μ , $\bar{x} = \frac{1}{q} \sum_{i=1}^q \bar{x}^{(i)}$ can be used. Efron and Morris (1972) used the following argument to estimate A .

Since $\bar{X}^{(i)}$ has the marginal distribution

$$\bar{X}^{(i)} \sim N(\mu, \frac{1}{\bar{N}} \Sigma + B) \quad (i=1, \dots, q)$$

the marginal distribution of

$$\tilde{S} = \sum_{i=1}^q (\bar{X}^{(i)} - \bar{X})(\bar{X}^{(i)} - \bar{X})' \sim W(\frac{1}{\bar{N}} \Sigma + B, q-1, k),$$

$$\text{where } \bar{X} = \frac{1}{q} \sum_{i=1}^q \bar{X}^{(i)}.$$

A standard calculation shows that

$$E \{ (q-k-2) \tilde{S}^{-1} \} = (\frac{1}{\bar{N}} \Sigma + B)^{-1}.$$

$(q-k-2) \{ \sum_{i=1}^q (\bar{X}^{(i)} - \bar{X})(\bar{X}^{(i)} - \bar{X})' \}^{-1}$ can therefore be used as an estimate of $(\frac{1}{\bar{N}} \Sigma + B)^{-1}$.

Since

$$\sum_{i=1}^q \sum_{j=1}^{\bar{N}} (X_j^{(i)} - \bar{X}^{(i)})(X_j^{(i)} - \bar{X}^{(i)})' \sim W(\Sigma, (\bar{N}-1)q, k)$$

and independently of $\tilde{S}$, Efron and Morris (1972) proposed

$$\frac{1}{\{(\tilde{N}-1)q + k + 1\}} \sum_{i=1}^q \sum_{j=1}^{\tilde{N}} (x_j^{(i)} - \bar{x}^{(i)}) (x_j^{(i)} - \bar{x}^{(i)})'$$

as an estimate of Σ .

$$(3.2) \text{ Let } \tilde{M}_1 = \frac{1}{q(\tilde{N}-1)} \sum_{i=1}^q \sum_{j=1}^{\tilde{N}} (x_j^{(i)} - \bar{x}^{(i)}) (x_j^{(i)} - \bar{x}^{(i)})'$$

and

$$(3.3) \quad \tilde{M}_2 = \frac{\tilde{N}}{q-1} \sum_{i=1}^q (\bar{x}^{(i)} - \bar{x}) (\bar{x}^{(i)} - \bar{x})',$$

then the Stein estimator of $m^{(i)}$ as derived by Efron and Morris (1972) is given by

$$(3.4) \quad \hat{A}^{(st)}_{\bar{x}} + (1 - \hat{A}^{(st)})_{\bar{x}} m^{(i)}$$

$$\text{where } \hat{A}^{(st)} = \frac{q-k-2}{(\tilde{N}-1)q+k+1} \frac{q(\tilde{N}-1)}{q-1} \tilde{M}_1 \tilde{M}_2^{-1}.$$

The risk of the Stein estimator is given by

$$(3.5) \quad \tilde{N} \sum_{i=1}^q (\hat{A}^{(st)}_{\bar{x}} + (1 - \hat{A}^{(st)})_{\bar{x}} m^{(i)})' \Sigma^{-1} (\hat{A}^{(st)}_{\bar{x}} + (1 - \hat{A}^{(st)})_{\bar{x}} m^{(i)}) \\ = pk - \frac{(\tilde{N}-1)q(q-k-2)^2}{(\tilde{N}-1)q+k+1} E \operatorname{tr} S^{-1} \Sigma$$

$$\leq pk, \text{ the risk of the maximum likelihood estimator } \bar{X}^{(i)}.$$

Thus conditionally on $m^{(i)}$ or unconditionally, the Stein estimator always outperforms the maximum likelihood estimator for estimating the means of normal populations if quadratic loss is used as our measure.

An even better estimator than (3.4) is the modified Stein estimator

$$(3.6) \quad \hat{A}_m^{(st)} \bar{x} + (1 - \hat{A}_m^{(st)})_{\bar{x}} m^{(i)},$$

where $\hat{A}_m^{(st)}$ is the expression for $\hat{A}^{(st)}$ modified so that all the roots are between 0 and 1 and can be obtained as follows:

$\hat{A}_m^{(st)} = \tilde{\Gamma} \Lambda \tilde{\Gamma}^{-1}$, where $\tilde{\Gamma}$ is a $k \times k$ nonsingular matrix and Λ a diagonal matrix with λ_{tt} equal to the t -th largest eigenvalue, $t=1, \dots, k$ and $\hat{A}_m^{(st)} = \tilde{\Gamma} \Lambda_m \tilde{\Gamma}^{-1}$, where Λ_m has diagonal elements $\min \{1, \lambda_{tt}\}$.

It can be shown that the risk of (3.6) is uniformly smaller than that of (3.4).

Efron and Morris (1972) used a computer simulation to estimate the exact size of Pearson's chi-square test for comparing binomial means. Comparisons of the various estimates with the true values revealed Stein's estimator and its multivariate generalizations (3.4) and (3.6) to be approximately twice as efficient as the maximum likelihood estimator. They also discussed the use of (3.6) for analyzing interactions in a two-way analysis of variance.

The purpose of this paragraph is to derive an alternative Stein type estimator (predictor) and to compare it with (3.6) as well as with the maximum likelihood estimator.

As mentioned by Geisser (1974), the predictive sample reuse (PSR) method is a useful tool for estimating unknown parameters. By using a PSR method the unknown A can be estimated as follows: Omit the j -th vector $(x_j^{(1)}, \dots, x_j^{(q)})$ from every population, using $(1-A)\bar{x}_j^{(i)} + A\bar{x}_j$ to predict it, where

(3.7) $\bar{x}_j^{(i)} = \frac{1}{\bar{N}-1} (\bar{N}\bar{x}^{(i)} - x_j^{(i)})$, i.e. the mean of the $\bar{N}-1$ observations from the i -th population omitting the j -th observation, and

$$(3.8) \quad \bar{x}_j = \frac{1}{q(\bar{N}-1)} (q\bar{N}\bar{x} - \sum_{i=1}^q x_j^{(i)}),$$

the overall mean if the j -th vector is omitted.

Repeat this for $j=1, \dots, \bar{N}$ and compute the mean squared prediction error,

$$(3.9) \quad V_A^2 = \frac{1}{\bar{N}q} \sum_{i=1}^q \sum_{j=1}^{\bar{N}} \{ (1-A)\bar{x}_j^{(i)} + A\bar{x}_j - x_j^{(i)} \}' \{ (1-A)\bar{x}_j^{(i)} + A\bar{x}_j - x_j^{(i)} \}.$$

This will be a sensible measure since the q populations have the same covariance matrix. Another measure that can be used is:

$$(3.10) \quad V_{A,\Sigma}^2 = \frac{1}{\bar{N}q} \sum_{i=1}^q \sum_{j=1}^{\bar{N}} \{ (1-A)\bar{x}_j^{(i)} + A\bar{x}_j - x_j^{(i)} \}' \Sigma^{-1} \{ (1-A)\bar{x}_j^{(i)} + A\bar{x}_j - x_j^{(i)} \}$$

Minimization of (3.9) or (3.10) with respect to A yields

$$(3.11) \quad A^* = \left\{ \sum_{i=1}^q \sum_{j=1}^{\bar{N}} (x_j^{(i)} - \bar{x}_j^{(i)}) (\bar{x}_j - \bar{x}_j^{(i)})' \right\} \left\{ \sum_{i=1}^q \sum_{j=1}^{\bar{N}} (\bar{x}_j - \bar{x}_j^{(i)}) (\bar{x}_j - \bar{x}_j^{(i)})' \right\}^{-1} \\ = \bar{N} (q\tilde{M}_1 - \tilde{M}_3) (q\tilde{M}_1 + (q-1)(\bar{N}-1)\tilde{M}_2 - \tilde{M}_3)^{-1}$$

where

$\tilde{M}_1$ and $\tilde{M}_2$ are defined as in equations (3.2) and (3.3) and

$$\tilde{M}_3 = \frac{q}{N-1} \sum_{j=1}^N (\bar{x}_{(j)} - \bar{x})(\bar{x}_{(j)} - \bar{x})'$$

$$\text{with } \bar{x}_{(j)} = \frac{1}{q} \sum_{i=1}^q x_j^{(i)}.$$

What we wish to consider is the simultaneous omission of q vectors one from each population in a nonarbitrary manner. This is accomplished by considering all possible permutations of observations within each sample and averaging V_A^2 over all such possibilities. It is clear that $\tilde{M}_1$ and $\tilde{M}_2$ are unaffected by such permutations and only $\tilde{M}_3$ need be averaged. This average can easily be shown to be $\tilde{M}_1$. Hence substitution in (3.11) yields

$$(3.12) \quad \hat{A} = N M_1 ((N-1) \tilde{M}_2 + \tilde{M}_1)^{-1}$$

Usually $\hat{A}_m$ is used as an estimate of A where $\hat{A}_m$ is the expression for $\hat{A}$, modified so that all the roots are between 0 and 1. The estimator

$$(3.13) \quad (I - \hat{A}_m) \bar{x}_j^{(i)} + \hat{A}_m \bar{x}$$

is called the PSR estimator.

A similar result was obtained by Geisser (1974) for the case $k=1$. The other results given by Geisser (1974) and (1975) can easily be extended for general k . If we consider the special case $\Sigma = \sigma^2 I(k \times k)$ and $B = b^2 I(k \times k)$ where σ^2 and b^2 are constants then equation (3.1) simplifies to

$$(3.14) \quad (1 - \alpha) \bar{x}^{(i)} + \alpha \bar{x}$$

where

$$\alpha = \frac{1}{N} \sigma^2 (b^2 + \frac{1}{N} \sigma^2)^{-1}.$$

By minimizing the mean squared prediction error (obtained from the PSR method) with respect to α and using the same arguments as before it is found that

$$(3.15) \quad \hat{\alpha} = N \tilde{m}_1 ((N-1) \tilde{m}_2 + \tilde{m}_1)^{-1},$$

where

$$\tilde{m}_1 = \frac{1}{q(N-1)} \sum_{i=1}^q \sum_{j=1}^N (x_j^{(i)} - \bar{x}^{(i)})(x_j^{(i)} - \bar{x}^{(i)})'$$

and

$$\tilde{m}_2 = \frac{N}{q-1} \sum_{i=1}^q (\bar{x}^{(i)} - \bar{x})' (\bar{x}^{(i)} - \bar{x})$$

It is however better to use $\hat{\alpha}_m$ which is $\min(\hat{\alpha}, 1)$ as an estimate of α in (3.13).
The estimator (predictor)

$$(3.16) \quad (1 - \hat{\alpha}_m) \bar{x}^{(i)} + \hat{\alpha}_m \bar{x}$$

is the analogue of the univariate predictor of Geisser (1975) and will be of best use if all the variables have the same variance and if all covariances are equal to zero.

Three examples were considered to compare the four predictors (estimators):

$$(3.17) \quad \psi_1^{(i)} = \bar{x}^{(i)}$$

$$(3.18) \quad \psi_2^{(i)} = \hat{\alpha}_m \bar{x} + (1 - \hat{\alpha}_m) \bar{x}^{(i)}$$

$$(3.19) \quad \psi_3^{(i)} = \hat{A}_m \bar{x} + (I - \hat{A}_m) \bar{x}^{(i)}$$

$$(3.20) \quad \psi_4^{(i)} = \hat{A}_m^{(st)} \bar{x} + (I - \hat{A}_m^{(st)}) \bar{x}^{(i)}$$

$$i=1, \dots, q.$$

In the first two examples $m^{(i)}$; $i=1, \dots, q$ was considered fixed and in the third example $m^{(i)}$ was considered random.

Example 3.1

Case I

Consider the $k=3$ dimensional normal population $N(\mu, B)$ where

$$\mu' = (4 \quad 9 \quad 0) \quad \text{and} \quad B = \begin{bmatrix} 1 & 4 & 1 \\ 4 & 20 & 8 \\ 1 & 8 & 6 \end{bmatrix}$$

from which $q = 6$ vectors were taken:

$$\begin{array}{ll} m^{(1)'} &= (3,1 \quad 9,6 \quad 2,1) & m^{(4)'} &= (3,8 \quad 4,5 \quad -1,4) \\ m^{(2)'} &= (4,7 \quad 10,8 \quad 0,7) & m^{(5)'} &= (6,6 \quad 19,9 \quad 3,4) \\ m^{(3)'} &= (5,3 \quad 13,9 \quad 0,9) & m^{(6)'} &= (3,3 \quad 5,0 \quad -1,5) \end{array}$$

From each of the normal populations $N(m^{(i)}, \Sigma)$, $i=1, \dots, q$,

where

$$\Sigma = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 8 \\ 3 & 8 & 14 \end{bmatrix}$$

a random sample of $\tilde{N}=30$ observations were drawn.

To illustrate the suitability of each of the four predictors; (3.17), (3.18), (3.19) and (3.20), for predicting a future observation $X_{\tilde{N}+1}^{(i)}$ from each of the populations with means $m^{(i)}$; $i=1, \dots, q$, an actual future observation was simulated and their estimated risks, using two separate quadratic loss functions

$$R_{\ell} = \frac{\tilde{N}}{(\tilde{N}+1)\tilde{S}} \sum_{i=1}^q \sum_{\ell=1}^4 \{ \psi_{\ell}^{(i)} - X_{\tilde{N}+1}^{(i)} \}' \Sigma^{-1} \{ \psi_{\ell}^{(i)} - X_{\tilde{N}+1}^{(i)} \} \quad \ell=1, 2, 3, 4$$

and

$$\tilde{R}_{\ell} = \frac{\tilde{N}}{(\tilde{N}+1)\tilde{S}} \sum_{i=1}^q \sum_{\ell=1}^4 \{ \psi_{\ell}^{(i)} - X_{\tilde{N}+1}^{(i)} \}' \{ \psi_{\ell}^{(i)} - X_{\tilde{N}+1}^{(i)} \}$$

were calculated. $\tilde{S} = 4000$ is the number of times the process was repeated. The results are given in the following table:

Predictor		ψ_1	ψ_2	ψ_3	ψ_4
Estimated risks	R	17,904	18,033	17,871	17,885
	$\tilde{R}$	119,045	119,022	118,410	118,704

where $\psi_{\ell} = (\psi_{\ell}^{(1)}, \dots, \psi_{\ell}^{(6)})$; $\ell=1, 2, 3, 4$.

Example 3.2

Case I

The same as example 3.1, Case 1, but with

$$\mu = (3 \ 4 \ 5), B = \begin{bmatrix} 0,1 & 0,4 & 0,1 \\ 0,4 & 2,0 & 0,8 \\ 0,1 & 0,8 & 0,6 \end{bmatrix}, \Sigma = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 8 \\ 3 & 8 & 14 \end{bmatrix}$$

$k=3, q=6$ and

$$m^{(1)'} = (2,9593 \quad 2,9149 \quad 4,2538)$$

$$m^{(2)'} = (3,0994 \quad 4,4066 \quad 5,3550)$$

$$m^{(3)'} = (3,3495 \quad 6,1094 \quad 6,0795)$$

$$\tilde{N} = 30 \text{ and } \tilde{S} = 4000$$

$$m^{(4)'} = (2,9850 \quad 4,1351 \quad 4,7802)$$

$$m^{(5)'} = (3,6467 \quad 7,0072 \quad 6,1840)$$

$$m^{(6)'} = (2,9558 \quad 3,8893 \quad 4,8184)$$

Predictor		ψ_1	ψ_2	ψ_3	ψ_4
Estimated	R	17,799	18,832	17,678	17,749
risks	$\tilde{R}$	118,947	118,626	117,168	118,284

Example 3.3

Case I

In this example μ' , B and Σ were taken the same as in the case of example 3.2 again $\tilde{N}=30$, but for each of the 4000 repetitions a new $m^{(i)}$ ($i=1, \dots, q$) was drawn. The risks obtained are therefore the actual Bayes risks. The results are given in the following table;

Predictor		ψ_1	ψ_2	ψ_3	ψ_4
Estimated	R	18,347	19,587	18,230	18,296
risks	$\tilde{R}$	123,229	122,904	121,818	122,618

The values 17, 904; 17, 799 and 18,347 of R_1 for the three examples compare well with

$$E(\tilde{N}+1)^{-1} \tilde{N} \sum_{i=1}^q \{ \psi_1^{(i)} - X_{\tilde{N}+1}^{(i)} \}' \Sigma^{-1} \{ \psi_1^{(i)} - X_{\tilde{N}+1}^{(i)} \} = 18$$

the exact value of the risk of the maximum likelihood predictor ψ_1 .

The pattern for the four predictors for the different examples are exactly the same.

It is clear that ψ_3 and ψ_4 are better predictors than ψ_1 conditionally on $m^{(i)}$ (examples 3.1 and 3.2) and unconditionally (example 3.3). This would have been the case for Stein predictor ψ_4 , but the most striking aspect however is that ψ_3 is better than ψ_4 in the three examples given. While this is no conclusive proof of the superiority of the PSR method, it does indicate that, under certain conditions, it is a worthy competitor for the Stein predictor.

From the tables it is also clear that ψ_3 and ψ_4 are better predictors than ψ_2 . This however is not surprising since the design of the experiment for the three examples favours predictors of the form $A\bar{x} + (I-A)\bar{x}^{(i)}$ rather than predictors of the form $\alpha\bar{x} + (1-\alpha)\bar{x}^{(i)}$ which also indicates that for suitable cases the matrix PSR predictor (ψ_4) will be better than the univariate predictor of Geisser (1975), applied co-ordinate-wise to the components of $x_j^{(i)}$ separately.

The three examples were also used to compare ψ_1, ψ_2, ψ_3 and ψ_4 as estimators of the population means $m^{(i)}, i=1, \dots, 6$. The estimated risks:

$$R_{\ell}^* = \frac{\tilde{N}}{\tilde{S}} \sum_{i=1}^q \{ \psi_{\ell}^{(i)} - m^{(i)} \}' \Sigma^{-1} \{ \psi_{\ell} - m^{(i)} \},$$

$$R_{\ell}^* = \frac{\tilde{N}}{\tilde{S}} \sum_{i=1}^q \{ \psi_{\ell}^{(i)} - m^{(i)} \}' \{ \psi_{\ell} - m^{(i)} \} \quad \ell=1, 2, 3, 4,$$

were calculated with $\tilde{S} = 4000$ and the following results were obtained.

Example 3.1

Case II

Estimator		ψ_1	ψ_2	ψ_3	ψ_4
Estimated risks	R^*	18,059	21,768	17,018	17,369
	$\tilde{R}^*$	119,740	118,811	98,566	106,678

Example 3.2

Case II

Estimator		ψ_1	ψ_2	ψ_3	ψ_4
Estimated risks	R^*	18,031	51,645	14,096	16,240
	$\tilde{R}^*$	120,631	112,950	63,656	96,859

Example 3.3

Case II

Estimator		ψ_1	ψ_2	ψ_3	ψ_4
Estimated	R^*	17,949	58,816	15,300	16,521
risks	$\tilde{R}^*$	117,532	104,946	70,915	96,064

The results for estimation show the same pattern as in the case of prediction except that the improvements of ψ_3 and ψ_4 over ψ_1 are much greater in the case of estimation than in the case of prediction. The reason for this is that in the case of the first quadratic loss function;

$$\text{Risk (estimation)} = (\tilde{N}+1) \text{Risk (prediction)} - kq\tilde{N}$$

and that for the second quadratic loss function;

$$\tilde{\text{Risk (estimation)}} = (\tilde{N}+1) \text{Risk (prediction)} - q\tilde{N}\text{tr}\Sigma,$$

which show that the relative differences between the risks become larger when ψ_ℓ ; $\ell=1, 2, 3$ are used as estimators instead of predictors.

For these examples the $m^{(i)}$ ($i=1, \dots, q$) were obtained by sampling from a normal population. This may not be necessary although the sample reuse procedure was motivated by such a consideration.

The reason being that the PSR method should be able to compete with the Stein procedure on the frequentist basis for which superiority is claimed for the Stein approach to standard maximum likelihood estimation. Although the PSR method here is based on a predictive function suggested by normality and other model considerations, it is anticipated that it could be highly robust to certain changes in distribution if the groups were essentially exchangeable except for location and because the method is to a degree data-adjusted.

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